Fiscal Year:	FY 2007	**Task Last Updated:**	FY 05/01/2007
PI Name:	Stowe, Raymond Ph.D.		
Project Title:	Effects of Simulated Spaceflight on Virus-Specific Immunity		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Operational and clinical research		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	77568-3318	**Congressional District:**	22
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	01-OBPR-07
Start Date:	09/01/2002	**End Date:**	12/31/2006
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Received NCE to 12/31/2006, per PI (9/29/06)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Pierson, Duane Ph.D. (JSC)		
Grant/Contract No.:	NNJ04HD75G		
Performance Goal No.:			
Performance Goal Text:			
	The long-term goals of this project are to understand the effects of spaceflight (i.e., altered gravitational forces, stress-associated changes, etc.) on viral immunity. Our group has studied neuroimmune function during short-term spaceflights. Elevated levels of cortisol were found after launch, an event involving hypergravity followed by microgravity. Significant increases in stress hormones (i.e., cortisol, epinephrine, norepinephrine) were also found immediately after landing (another event involving hypergravity). Importantly, we found evidence of decreased cellular immunity and increased reactivation of latent herpesviruses in astronauts during and after spaceflight. Opportunistic infections (e.g., latent herpesviruses) pose an important health risk to crewmembers during spaceflight, and this risk will most certainly increase during long-term spaceflights. Lack of a suitable ground-based model that simulates the multiple gravitational changes that occur during spaceflight (i.e., hypergravity at launch, microgravity during flight,		

Task Description:	hypergravity at landing) has limited investigations on how acute gravitational changes affect the immune system. To overcome this limitation, human subjects will undergo acute acceleration (hypergravity) and bedrest to simulate certain conditions that occur in space. This model system will be used to test the hypothesis that simulated spaceflight and associated stress will decrease virus-specific cellular immunity and reactivate latent herpesviruses. Neuroendocrine hormone levels, immune responses, and latent herpesvirus reactivation will be compared between test subjects and control subjects. The results of this study will lead to a better understanding of the physiological changes associated with spaceflight, and how these changes may result in altered immunity and latent virus reactivation. These data will subsequently be used to develop a mechanistic investigation on how neuroendocrine hormones modulate virus-specific immunity in the spaceflight environment.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This experiment will address fundamental questions on stress, spaceflight, and virus-specific immunity and lead to countermeasures for exploration class missions. These countermeasures may be applicable to clinical populations such as post-transplant or AIDS patients.
Task Progress:	Twelve subjects, screened at the Johnson Space Center (JSC) Test Subject Facility, were recruited for four bedrest campaigns. All subjects were seropositive for Epstein-Barr virus (EBV) and Cytomegalovirus (CMV) indicating past exposure. Blood, urine, and saliva were collected from subjects ten and four days before simulated launch (L-10; L-4). At L-1, subjects were transported to Brooks Air Force Base where they began daily collection of saliva and urine samples throughout the 16 days of bedrest. On “launch” day, blood was collected just prior to centrifugation as well as 15 and 240 minutes after centrifugation. Blood was also collected at bedrest (BR) + 4 days, BR+8, and BR+12. On “landing” day, blood was again collected in the same fashion as launch day. Subjects were then transported back to JSC and released. Post-BR samples were collected at recovery (R)+2 and R+15 days. Acute increases in cortisol were found on launch day; cortisol was also increased on landing day. Total white blood cell (WBC) counts were elevated immediately after simulated launch, primarily due to demargination of polymorphonuclear leukocytes (neutrophils). WBC counts returned to baseline levels by BR+4. At simulated landing, WBC counts were again elevated but at a greater magnitude. Salivary cortisol levels were elevated after centrifugation and generally mirrored the urinary cortisol levels; EBV shedding in saliva was also increased during stressful periods. Flow cytometric analysis indicated significant changes occurred within leukocyte and lymphocyte subpopulations. Intracellular cytokine production by SEB-stimulated CD4+ T cells decreased after centrifugation. The percentages of CMV- and EBV-specific CD 8+ T cells were elevated in some subjects although their function (e.g., intracellular cytokine production) was decreased and correlated with increased stress hormones. The pattern of increased stress hormones (i.e., cortisol and catecholamines) from our test subjects closely matches the stress hormone data from short-term space flights. In addition, further similarities were noted in the shifts of leukocyte and lymphocyte subsets. These data support the hypothesis that this model mimics certain stresses that occur during space flight. We conclude that we have developed a novel model of space flight, and this model will provide a testing ground for pharmacological and behavioral strategies designed to mitigate changes associated with space flight.
Bibliography Type:	Description: (Last Updated: 03/07/2019)
Articles in Peer-reviewed Journals	Pierson DL, Stowe RP, Phillips TM, Lugg DJ, Mehta SK. "Epstein-Barr virus shedding by astronauts during space flight." Brain Behav Immun. 2005 May;19(3):235-42. PMID: 15797312 , May-2005
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Articles in Peer-reviewed Journals	Ruiz RJ, Stowe RP, Goluszko E, Clark MC, Tan A. "The relationships among acculturation, body mass index, depression, and interleukin 1-receptor antagonist in Hispanic pregnant women." Ethn Dis. 2007 Spring;17(2):338-43. PMID: 17682368 , Apr-2007
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Books/Book Chapters	Pierson DL, Mehta SK, Stowe RP. "Reactivation of Latent Herpes Viruses in Astronauts." in "Psychoneuroimmunology, Fourth Edition." Ed. R. Ader. Boston : Academic Press, 2007. p. 851-868., Jan-2007